



Chapter 4

Laws of Motion

PHYSICS

CLASS - 11

NCERT BASED

For NEET UG Aspirants

The motion of a body is governed by certain fundamental principles known as the **Laws of Motion**.

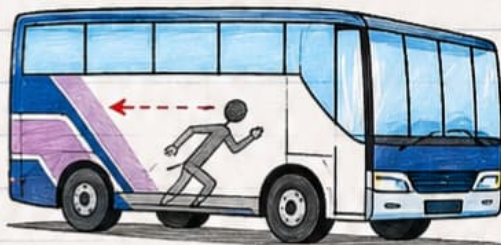
1. Inertia

→ Inertia is the property of a body due to which it resists any change in its state of rest or of uniform motion in a straight line.

→ Inertia is of two types -

- Inertia of rest**: Tendency of a body to remain at rest.
- Inertia of motion**: Tendency of a body to continue in its state of uniform motion in a straight line.

Example



When the bus suddenly stops, the body tends to keep moving forward due to inertia of motion.

Newton's Three Laws of Motion

1st Law

"A body continues to remain in its state of rest or of uniform motion in a straight line unless acted upon by an external unbalanced force."

- It is also called the law of inertia.
- It defines the natural state of a body.



Examples

- ★ A book placed on a table remains at rest.
- ★ A hockey puck sliding on ice continues to move in a straight line with constant speed.

2nd Law

"The rate of change of momentum of a body is directly proportional to the net external force acting on it and takes place in the direction of the force."

$$F \propto \frac{dp}{dt}$$

$$\text{or, } F = k \frac{dp}{dt}$$

(where k is constant)

In SI units, $k = 1$

$$F = \frac{dp}{dt}$$

where,

F = net external force

p = momentum ($p = mv$)

t = time



motion in direction of force



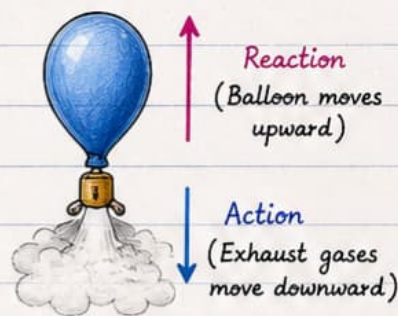
More force
→ more acceleration

3rd Law

"To every action, there is an equal and opposite reaction."

- The two forces act on two different bodies.
- They are equal in magnitude, opposite in direction and act along the same straight line.
- They never cancel each other because they act on different bodies.

Example



Types of Inertia

- ★ **Inertia of rest**: A body at rest tends to remain at rest.
- ★ **Inertia of motion**: A body in motion tends to continue in its state of motion.
- ★ **Inertia of direction**: A body moving in a particular direction tends to continue in the same direction.

Common Example

When a bus starts suddenly, passengers jerk backward (because of inertia of rest). When it stops suddenly, passengers jerk forward (because of inertia of motion).



Body tends to remain at rest



Body tends to continue in its motion

Some Important Points

- ① Force is a **vector** quantity.
- ② If the **resultant (net) external force** on a body is zero, its momentum remains constant.
- ③ More the mass of a body, more is its inertia.
- ④ Force can change the state of rest or uniform motion or direction of a body.

Remember!

Inertia is a **property** of matter while mass is a **measure** of the amount of matter.

Real Life Applications

1. Seat belt in car

Helps to reduce the effect of inertia when the car stops suddenly.



2. Hammer and wooden handle

When the wooden handle is hit sharply on the ground, the head of the hammer moves downward due to inertia of motion.



3. Dusting of carpet

Beating a carpet with a stick makes the dust particles come out due to inertia of rest.



In Short

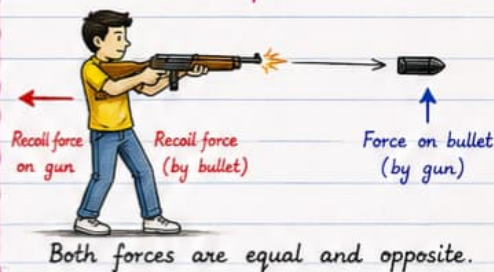
Newton's three laws of motion form the basis of classical mechanics. They explain how bodies behave under the action of forces and help us understand many everyday phenomena.

4th Law

"To every action, there is an equal and opposite reaction."

- The forces always occur in **pairs**.
- They act on two **different bodies**.
- They are **equal** in magnitude but **opposite** in direction.

Example



Friction

Friction is a force that **resists** the relative motion between two surfaces in contact.

It always acts **opposite** to the direction of motion.

Types of Friction

- ★ 1. Static friction
- ★ 2. Kinetic (sliding) friction
- ★ 3. Rolling friction
- ★ 4. Fluid friction

1. Static friction

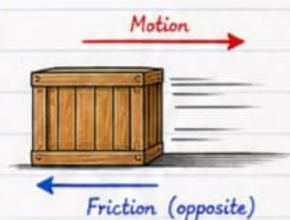
The friction which comes into play when one body tends to move over another but does not actually move.



Body remains at rest

2. Kinetic (sliding) friction

The friction which acts when one body slides over the surface of another body.



3. Rolling friction

The friction which acts when one body rolls over the surface of another body.



4. Fluid friction

The friction experienced by a body when it moves through a liquid or a gas.



Some Important Points

- Friction is a **contact** force.
- It depends on the **nature** of surfaces in contact.
- It is **independent** of the area of contact.
- It does not depend on the speed of motion.
- In case of rolling friction, it is much smaller than sliding friction.
- Friction can be both **useful** and **harmful**.

Useful Effects

- ✓ Helps in walking without slipping.
- ✓ Tyres get grip on the road.
- ✓ Eraser removes pencil marks.
- ✓ Brakes stop the moving vehicles.
- ✓ We can write with a pencil or pen.



Harmful Effects

- ✗ Causes wear and tear of machine parts.
- ✗ Producing heat and energy loss.
- ✗ Reduces efficiency of machines.



Newton's Laws in Daily Life

1. Inertia of rest : When we pull the table cloth quickly, the dishes on it remain at rest.



2. Inertia of motion : When a car stops suddenly, passengers tend to move forward.



3. Action-reaction : When we jump from a boat onto the shore, the boat moves backward.



Remember !

Newton's laws are valid in inertial frames of reference.

In Short

- 1st Law → Inertia
- 2nd Law → $F = \frac{dp}{dt}$
- 3rd Law → Action - Reaction
- 4th Law → Equal & opposite

Key Formula

$$F = \frac{dp}{dt}$$

$$= m \frac{dv}{dt}$$

$$= ma$$

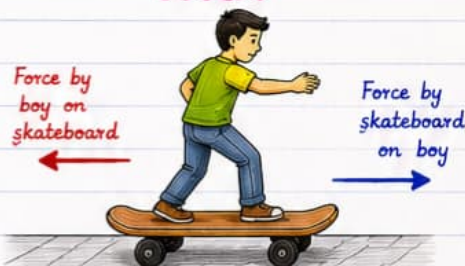
Keep Learning

5th Law

★ "The mutual forces of action and reaction between two bodies are always equal and opposite in direction."

- These forces act on **different bodies**, so they never cancel each other.
- They occur in **pairs**.

Example



Both forces are equal and opposite but act on **different bodies**.

Some Important Concepts

- ★ Force is a **vector** and has both magnitude and direction.
- ★ **Balanced forces** : Net force is zero, body remains in rest or uniform motion.
- ★ **Unbalanced forces** : Net force is non-zero, body changes its state of motion.
- ★ **Mass** is a measure of inertia.
- ★ **Momentum** of a body, $p = mv$ (vector quantity)
- ★ More the mass, more the inertia and more the force required to change its state.

Remember !

Force is not always needed to keep a body in motion. Only an unbalanced force is needed to **change** the state of motion.

Common Examples

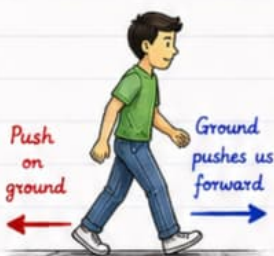
① Pushing a Wall

You push the wall but it does not move. The wall also pushes you with equal force in opposite direction.



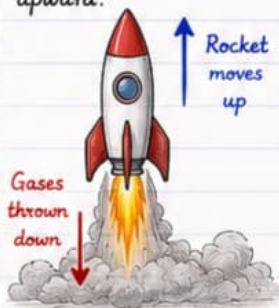
② Walking

We push the ground backward. The ground pushes us forward.



③ Rocket Launch

Hot gases are thrown downward. Rocket moves upward.



④ Swimming

Swimmer pushes water backward. Water pushes swimmer forward.



Quick Revision - Newton's Laws

1 st Law	Law of inertia. A body remains in rest or in uniform motion in a straight line unless acted upon by an external unbalanced force.
2 nd Law	Net force = rate of change of momentum. $F = \frac{dp}{dt}$ (For constant mass, $F = ma$)
3 rd Law	For every action, there is an equal and opposite reaction.
4 th Law	Forces act in pairs on different bodies. They are equal and opposite but do not cancel.
5 th Law	Mutual forces of action and reaction are equal and opposite in direction.

→ Understand - Apply - Excel !

Key Formulas

• Momentum : $p = mv$

• Newton's 2nd Law : $F = \frac{dp}{dt}$

For constant mass :

$F = ma$

• Units :

- Force - newton (N)
- Mass - kilogram (kg)
- Momentum - kg m s^{-1}
- Time - second (s)

Dimensional formula

$[F] = \text{MLT}^{-2}$

Nature is governed by simple laws.

Understand them deeply, and you can solve the **mysteries of motion**.